Software Requirements Specification (SRS)

Project Title: LokVeda — Your E-Gram Panchayat

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Version: v1.0.0

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1. Introduction

1.1 Purpose

This document describes the Software Requirements Specification (SRS) for **LokVeda — Your E-Gram Panchayat**, an academic web-based application developed to digitally represent core functionalities of a village-level governance system.

The purpose of this document is to provide a clear understanding of the system's functionality, design constraints, and future scope for academic evaluation and documentation.

1.2 Scope

LokVeda is a web-based platform designed to simulate an **e-governance system** for Gram Panchayat operations.

The current implementation focuses on:

- Secure authentication
- Session management
- Role-based access (User, Staff, Admin)
- UI/UX interaction
- Location validation
- Modular frontend architecture

Advanced backend operations and real-world service processing are intentionally kept for **future scope**.

1.3 Definitions, Acronyms, and Abbreviations

- **OTP**: One Time Password
- **UI/UX**: User Interface / User Experience
- **SRS**: Software Requirements Specification
- **Admin**: Panchayat authority
- **Staff**: Panchayat staff member
- **User**: Registered citizen

1.4 References

- IEEE SRS Documentation Standard
- Firebase Documentation
- HTML5, CSS3, JavaScript (MDN Web Docs)

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2. Overall Description

2.1 Product Perspective

LokVeda is a **standalone web application** developed using frontend technologies and Firebase as a backend service simulator.

Firebase is treated as a **virtual backend**, where authentication, database access, and session validation are encapsulated within a single module (`firebase.js`).

2.2 Product Functions

The system provides the following high-level functions:

- Role-based login (User / Staff / Admin)
- OTP-based authentication
- Secure session creation and validation
- Automatic session timeout handling
- Location-based access verification
- Greeting and personalization features
- Modular navigation system
- Dark mode support
- Logout safety mechanisms

2.3 User Classes and Characteristics

User Type	Description
User	Citizen registered in the village database
Staff	Panchayat staff with regional permissions
Admin	Panchayat administrator with national-level access

2.4 Operating Environment

- Web Browser (Chrome, Firefox, Edge)
- Desktop and Mobile devices
- Internet connection
- Firebase backend services

2.5 Design and Implementation Constraints

- Frontend-only implementation
- Firebase used to mimic backend behavior
- No real government database integration
- Academic usage only

2.6 Assumptions and Dependencies

- Users are pre-registered in the database
- OTP delivery is assumed to be reliable
- Browser supports modern JavaScript (ES Modules)

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3. System Features

3.1 Authentication System

- Aadhaar-based identity verification
- OTP verification for secure login
- Regex validation for input security
- User-friendly status messages

3.2 Session Management

- Secure session creation on login
- Automatic logout on inactivity (15 minutes)
- Session invalidation on improper logout
- Login attempt tracking and cooldown mechanism

3.3 Role-Based Access Control

- Separate dashboards for User, Staff, and Admin
- Unauthorized access redirection
- Session guard to prevent direct URL access

3.4 Location Checker

- User access restricted to village boundaries
- Staff access restricted to state boundaries
- Admin access restricted within India
- Border-level accuracy support

3.5 Navigation and UI Module

- Dynamic navigation bar post-authentication
- Conditional visibility of navigation elements
- Search UI (functional design, logic pending)
- Logout confirmation modal

3.6 Greeting Bot

- Displays contextual greeting messages
- Personalized based on user role and name
- Time-based greeting logic

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4. External Interface Requirements

4.1 User Interface

- Clean and minimal UI
- Responsive design for different screen sizes
- Dark mode with user preference persistence
- Accessibility-friendly labels and prompts

4.2 Hardware Interfaces

Not applicable (Web-based application).

4.3 Software Interfaces

- Firebase Authentication
- Firebase Firestore Database
- Browser Local Storage

4.4 Communication Interfaces

- HTTPS-based communication with Firebase services

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5. Non-Functional Requirements

5.1 Performance

- Fast page load time
- Lightweight frontend modules

5.2 Security

- OTP-based authentication
- Session validation before page access
- Prevention of unauthorized data access

5.3 Reliability

- Session recovery handling
- Controlled logout behavior
- Error-handled authentication flows

5.4 Usability

- Clear feedback messages
- Minimal learning curve
- User-friendly navigation

5.5 Maintainability

- Modular JavaScript architecture
- Encapsulated backend logic
- Easy feature extension

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6. Future Scope

- Real backend implementation
- Service application submission and tracking
- Admin approval workflows
- Document upload and download (PDF/DOCX)
- Notification system
- Analytics dashboard
- Multi-language support

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7. Conclusion

LokVeda serves as a structured academic demonstration of an e-governance platform with a strong focus on authentication, session management, and modular frontend architecture. The system is scalable and designed for future expansion.
